

Weekly Problem 2

Ravi Kini

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Exercise 1

Part (a)

1	$\neg p$	
2	$p \wedge q$	
3	p	$\wedge E2$
4	$\neg\neg p$	$\neg\neg I3$
5	$\neg\neg p \vee q$	$\vee I4$
6	$\neg p \Rightarrow q$	mcI5
7	$(p \wedge q) \Rightarrow (\neg p \Rightarrow q)$	$\Rightarrow I2-6$

We then see that $\neg p \vdash (p \wedge q) \Rightarrow (\neg p \Rightarrow q)$ is a valid argument in Strong Kleene logic.

Part (b)

1	$p \Rightarrow q$	
2	$\neg p \Rightarrow q$	
3	$q \vee \neg q$	
4	q	
5	q	R4
6	$\neg q$	
7	$\neg p \vee q$	mcE1
8	$\neg p$	
9	q	$\Rightarrow$ E2, 8
10	q	
11	q	R11
12	q	$\vee$ E7, 8–9, 10–11
13	q	$\vee$ E3, 4–6, 7–13

We then see that $p \Rightarrow q, \neg p \Rightarrow q, q \vee \neg q \vdash q$ is a valid argument in Strong Kleene logic.